

ImpAct-Climate Micro-grants projects

A Best Practices Guide for Canadian Colleges and Institutes

Volume IV

COLLEGES & INSTITUTES CANADA

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SUSTAINABLE DEVELOPMENT GOALS 13 CLIMATE ACTION

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ImpAct-Climate is a five-year pan-Canadian project delivered by Colleges and Institutes Canada and funded by Environment and Climate Change Canada.

About Colleges and Institutes Canada

CICan is the national and international voice of Canada's largest post-secondary education network. It advocates, builds capacity, and drives knowledge to strengthen Canada's publicly supported colleges, institutes, CEGEPs, and polytechnics to meet Canada's biggest challenges. With more than **95%** of Canadians living within **50 km** of a member institution, and thanks to its extensive reach around the globe, CICan works to future-proof communities in Canada and abroad.

We respectfully acknowledge that CICan's offices in Ottawa are located on the traditional and unceded territory of the Algonquin Anishinaabe Nation.

This report was produced with the support of the Social Innovation Lab at Algonquin College and authored by Anna Nikiforova, a recent graduate of Algonquin College.

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Campus Living Labs

A Best Practices Guide for Canadian Colleges and Institutes

Volume IV

This fourth edition of the Campus Living Labs report highlights best practices from the fourth cohort of projects implemented during 2025–2026. These initiatives were supported by micro-grants, which enabled project development and execution throughout this period.

This report aims to provide a practical path for college and institute leadership, sustainability staff, and funders to support GHG reductions and environmental awareness.

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About ImpAct-Climate

CICan's ImpAct-Climate initiative raises awareness about greenhouse gases and encourages behaviour change to reduce emissions in the college and institute sector.

We connect and mobilize college and institute leaders, sharing the knowledge, skills, and tools needed to combat climate change and its impacts.

By using the United Nations Sustainable Development Goals (SDGs) as a framework for collaboration, we give colleges and institutes the opportunity to work together towards a common goal, to learn from each other, and maximize the impact of Canada's largest post-secondary network.



What is Micro-Grant?

Some important title about microgrants

As part of the final year of the ImpAct-Climate project, CICan launched the Micro-Grant initiative to maximize impact and extend the reach of project outcomes across the college and institute sector. Through this initiative, participating institutions received up to $8,000 to design and deliver campus-based activities focused on raising awareness, encouraging behaviour change, and contributing to greenhouse gas (GHG) emissions reduction.

The Micro-Grants were designed to enable rapid, practical action by empowering institutions to test ideas, engage students and staff, and pilot solutions that could be scaled or adapted over time. Projects were required to include a strong awareness component, demonstrate potential for GHG reduction, and incorporate community engagement.

Positioned as a targeted “top-up” in the project’s final phase, the initiative helped activate new participants while reinforcing momentum among engaged institutions. By supporting locally driven initiatives within a national framework, the Micro-Grant program fostered collaboration, knowledge sharing, and innovation, contributing to broader climate goals.

Microgrant Themes:



Indigenous Knowledges and Practices

Projects in this category focus on integrating Indigenous knowledge and practices within a climate change context, or are either led or co-led by Indigenous staff, faculty, students, or communities.

Did you know? Today, First Nations, Métis, and Inuit entities are partners or beneficiaries of almost 20 per cent of Canada's electricity-generating infrastructure, and almost all of that infrastructure is producing renewable energy.



Waste Management

Projects in this category focus on addressing waste management at any point of an institution's waste stream, such as cafeteria food or packaging waste, medical faculty waste, laboratory waste, administration-based waste, or on-campus retail.

Did you know? Globally, landfills are the third highest source of methane (CH4), primarily due to food waste10, in fact, 10%of our global emissions come from food we don't eat.



Food

Projects in this category focus on food consumed, grown, or sold on campus, including in campus cafes or restaurants or through culinary or agricultural programs.

Did you know? Food production is responsible for approximately 26%of global GHG emissions?

1 - Blah-blah (2002)
2 - Blah (1999)
3 - Blah (3321)



Green Buildings and Land Management

Projects in this category focus on use applications, approaches, or technologies to reduce or sequester GHGs as they pertain to buildings, infrastructure, or campus lands.

Did you know? Buildings represent 39%of global greenhouse gas emissions12, including 28%in operational emissions and 11%in building materials and construction13, so it's important to not only address how we design and run buildings, but what materials we use14.

Aligned with broader project goals, projects were asked to identify a primary theme from one of five categories, though many initiatives span multiple themes — an approach that was encouraged.

1 - Blah-blah (2002)
2 - Blah (1999)
3 - Blah (3321)

Indigenous Impact

The ImpAct-Climate project recognized that meaningful climate action must be grounded in diverse ways of knowing, including Indigenous perspectives, leadership, and approaches to sustainability.

Through the Micro-Grant initiative and broader project activities, participating colleges and institutes were encouraged to meaningfully engage Indigenous communities, knowledge holders, and learners in the design and delivery of climate initiatives. This included incorporating Indigenous knowledge systems, supporting participation of Elders and Knowledge Keepers, and creating space for community-led dialogue and learning.

These efforts helped strengthen relationships between institutions and Indigenous communities while reinforcing the importance of holistic, place-based approaches to climate action.

Algonquin College transformed the DARE District into a climate awareness experience that integrated Indigenous traditional knowledge from local Elders and Inuk entrepreneurs to foster a deeper community responsibility toward the land.





BCIT integrated Indigenous perspectives on environmental stewardship through educational videos featuring Indigenous staff to deepen the community's understanding of sustainable energy practices and collective responsibility toward respecting the Earth.



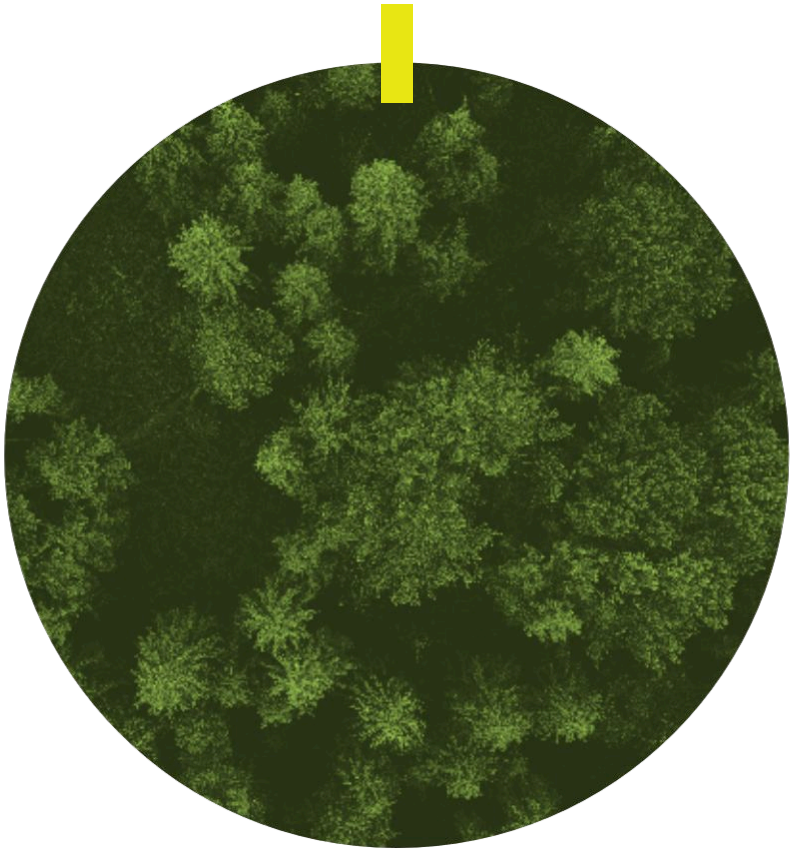
Collège Boréal utilized a new food waste dehydration system to test the growth of Indigenous medicinal plants in partnership with the First Nations, Métis, and Inuit Education Centre to enhance land stewardship.

Durham College integrated Indigenous knowledge systems through the creation of an Indigenous Healing Garden and land-based learning ceremonies to foster a community-wide responsibility toward Reconciliation and sustainable urban agriculture.





The University College of the North established a vermicomposting program and mural project at The Pas campus to emphasize the shared Indigenous stewardship role for wildlife, soil, and water through community-led environmental research.



Appendix:

Micro-Grants Projects

(2025-2026)

Creation of an Urban Mini-Forest (Miyawaki Method)

Collège La Cité

Project Lead: Ms. Lan Chi Nguyen Weekes
Collège La Cité, Ottawa, ON

LA CITÉ
LE COLLÈGE D'ARTS APPLIQUÉS ET DE TECHNOLOGIE

Project Description:

A 200 m² dense micro-forest was created on campus using the Miyawaki method, a rapid reforestation technique that involves planting native trees and shrubs very close together to mimic natural forest ecosystems. This method accelerates growth, improves biodiversity, and drastically increases carbon sequestration compared to conventional planting.



Key Highlights:

- **200 m² micro-forest** – a fully transformed campus plot now supporting a dense, thriving ecosystem that restores biodiversity in an urban environment.
- **600+ native trees and shrubs** – carefully selected and planted close together to replicate natural forest growth, creating a self-sustaining ecosystem.
- **3-10× faster growth** – achieved through the Miyawaki method, enabling rapid establishment and maturation compared to conventional planting techniques.
- **30× more effective** carbon capture – significantly reduces greenhouse gas emissions, demonstrating the grant's direct climate impact.
- **−1.3°C local cooling** – measurable reduction in urban heat island effect, improving the campus environment for students and staff.



The La Cité project proved the Miyawaki method's impact on biodiversity and GHG reduction, highlighting the importance of proper planning and monitoring. Next, the team aims to scale the forest, expand ecological monitoring, and leverage additional funding to maximize environmental and educational outcomes.





Field-to-fork climate action:

Strengthening Circular Food Systems through Indigenous Collaboration, Education and Knowledge Exchange

Durham College

Project Lead: Kelly O'Brien
Durham College, Oshawa, ON

BARRETT CENTRE | DURHAM COLLEGE

Project Description:



This project supports the integration of Indigenous perspectives within Durham College's Field-to-Fork-to-Field campus model and reflects our ongoing responsibility to engage in meaningful work toward Reconciliation. In partnership with the Barrett Centre of Innovation in Sustainable Urban Agriculture, students gain hands-on experience in GHG-reduction strategies through culinary programming, applied research, and industry-informed workshops.

Indigenous knowledge systems and voices are centred with the development of an Indigenous Healing Garden and integrated throughout learning, including garden design and creation in partnership with Miinikaan and a ground blessing ceremony led by Elder Dorothy Taylor. It invites our community to learn through relationships with the land and with the Indigenous knowledge systems and communities for sustainability. Community events and GHG-conscious culinary workshops model sustainable food preparation, composting, and waste-reduction practices, supported by biodigester and food- waste data shared through public outreach.

Key Highlights:

- **6.335 metric tonnes** – diverted to the campus biodigester, reducing food waste transported off campus by 30% and supporting a closed-loop, low-carbon food system.
- **2,000+ kg** – recaptured culinary lab by-products for value-added and future use within academic programming. For example, vegetable peels and scraps are dehydrated or freeze-dried to produce vegetable stock for future culinary labs, reducing waste while enhancing learning outcomes.
- **3,996 kg** – donated fresh produce to community partners since 2023, directly addressing food insecurity while reinforcing the connection between education, production, and community well-being.
- **3,604 community members** have participated in workshops, tours, and events, with a total of **87 workshops** delivered, expanding the reach of sustainable food systems education.



In collaboration with Durham College's First Peoples Indigenous Centre, Indigenous Education Circle, Miinikaan, Alumni Association, and the campus community, an Indigenous Agroecology Garden and raised-bed planters featuring medicinal plants will be installed on campus. Guided by Indigenous knowledge systems, this space will support curriculum integration and experiential learning.





Innovative Fertilization and Respect for Traditions:
Cultivation of Indigenous Medicinal Plants

Collège Boréal

Project Lead: Olivia Baudet
Collège Boréal, Sudbury, ON

Boréal

Project Description:



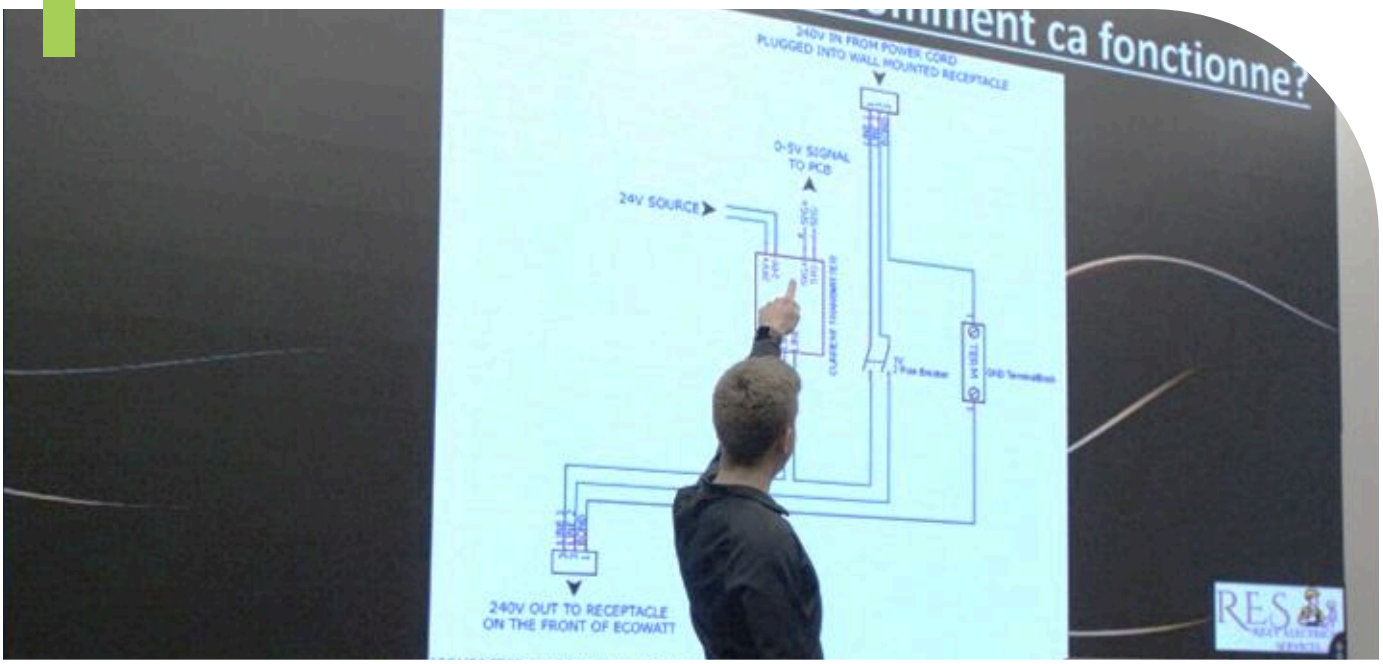
Each year, the campus cafeteria generates a significant amount of food waste which, when sent to landfill, contributes to greenhouse gas emissions. Through ImpAct Climate microfunding, a dehydrator was acquired in 2024 to divert this waste from landfill, then grind and dry it.

The main objective of this phase of the project is to reduce the campus carbon footprint by diverting food waste from landfill, while enhancing its agronomic potential. More specifically, the project aims to test the effectiveness of the dry substrate produced by the dehydrator on the growth of Indigenous medicinal plants, cultivated in partnership with the First Nations, Métis and Inuit Education Centre at Collège Boréal.

The project also includes the organization of awareness workshops open to the community, in collaboration with the City of Greater Sudbury and the Junction Creek Stewardship Committee (JCSC). These activities aim to promote waste sorting and the reduction of food waste, as well as to better understand the impact of waste on the local environment.

Key Highlights:

- **Nearly 450 kg** – of food waste diverted from landfill, preventing the emission of approximately 15 kg of methane and 440 kg of CO_2 into the atmosphere.
- **53 lb** – of waste successfully removed from the natural areas surrounding the campus.
- Strengthening of relationships and collaborations established with several local Indigenous communities.
- **Nearly 100 attendees** – attended the awareness day, primarily from the student community.
- **Approximately 30 participants** – took part in the environmental quiz activity.



The Collège Boréal plans to continue collecting food waste in order to reduce the quantity of materials sent to landfill. It also intends to continue organizing awareness activities and experimenting with different ways of using the substrate produced by the food dehydrator.





Shut the Sash Lambton
Lambton College

Project Lead: Carlos Diaz
Lambton College, Sarnia, ON

Project Description:



Shut the Sash Lambton is a pilot sustainability initiative at Lambton College designed to reduce laboratory energy use and greenhouse gas (GHG) emissions while reinforcing safe laboratory practices. The project couples sash-position sensors on selected research fume hoods with a targeted awareness and behaviour-change campaign that encourages lab users to close fume hood sashes when not actively in use. Because laboratory fume hoods are among the most energy-intensive systems in research facilities, this simple behaviour change can significantly reduce ventilation energy demand without impacting research productivity.

The initiative integrates applied research, facilities operations, and student engagement to establish a scalable model for safer, lower-carbon laboratory operations.

Key Highlights:

- **~20–40% reduction** – targeted decrease in ventilation energy demand in participating labs through the Lambton pilot, with greater savings achievable through campus-wide adoption.
- **77+ community members** – actively engaged in the campaign through in-lab orientations, posters and decals, digital communications, and a short educational video emphasizing both safety and sustainability benefits.
- The project reinforced the role of everyday operational habits as effective engineering controls, helping normalize sustainable laboratory practices while maintaining proper airflow and containment performance.
- The initiative demonstrates how simple, behaviour-based interventions can deliver meaningful energy and emissions reductions using low-cost technology and existing infrastructure, avoiding the need for expensive mechanical retrofits



Lambton College project will continue the pilot part of ongoing laboratory sustainability efforts at Lambton College. With new air handling units installed, the project team will continue collecting data to quantify energy, cost, and GHG impacts. Next steps include completing data analysis, producing a report documenting performance and lessons learned, and using the results to support scaling the program to additional laboratories across campus.



Cause a Scene
Algonquin College

Project Lead: Matthew Jerabek
Algonquin College, Ottawa, ON

ALGONQUIN COLLEGE

Project Description:



Cause a Scene transformed Algonquin College's DARE District into a public-facing climate awareness experience through art and interactive technology. Inspired by the DARE District's three-storey Indigenous mural, the initiative launched on Nov. 13, 2025 with a multigenerational climate dialogue and featuring Chuck Commanda (Indigenous Knowledge Keeper), Vanessa Stevens (Mamidosewin Centre), and Amenda Soucy (Inuk student entrepreneur, Business Management and Entrepreneurship). The dialogue invited employees, learners, and community members to reflect on responsibilities to land and future generations.

Following the dialogue, Illustration and Concept Art learners created nine large-scale murals and interactive activations across the first and second floors. The project was delivered through collaboration between the Social Innovation Lab and the Algonquin College Makerspace Innovation team, further engaging students from the Business Management and Entrepreneurship program and the Interactive Media Design Program to capture media content from the day, creating a highlight video and a virtual gallery of the artwork.

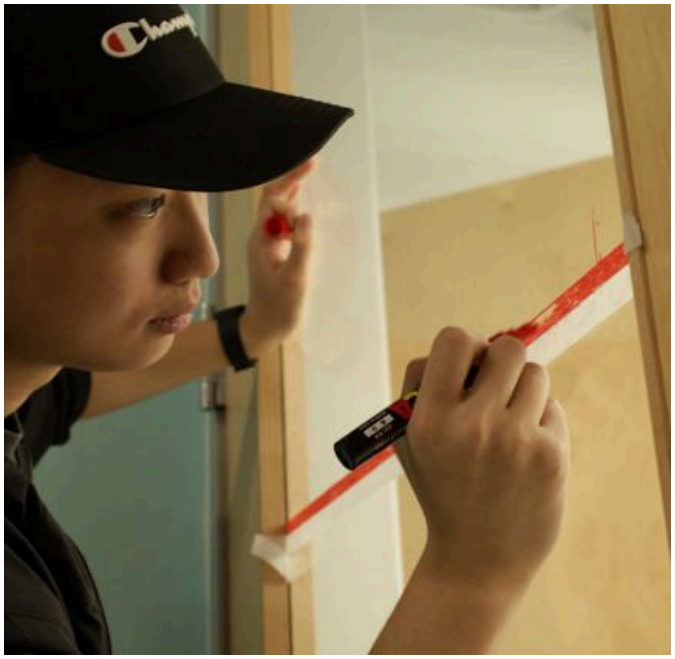
Key Highlights:

- **55 Illustration and Concept Art learners** engaged in a portfolio-ready class project centered on climate action and Indigenous perspectives.
- **9 murals installed** in a high-traffic campus corridor, on display from Nov. 13, 2025 to Feb. 12, 2026.
- **77 external visitors** – reached during the Career Fair period while the murals were on display.
- **2,141 unique students** – registered during the Career Fair period.
- **Approximately 730 prospective students and guests** – exposed to the murals during the display period, including 470 high school students and 260 visitors through general campus tours.
- **4 City of Ottawa attendees** – participated in a cohort visit to view the finalized murals, alongside in-person visits from CICan and Environment and Climate Change Canada representatives.



At Algonquin College, the mural initiative will continue within the Illustration and Concept Art program to extend its impact and sustain climate awareness. A second cohort of students has already been introduced to the project model and associated technology, supporting ongoing experiential learning and future activations.





Energy Awareness Week and Sweater Day
British Columbia Institute of Technology

Project Leads: Henry Leung, Stella Zhang, Green Team members
British Columbia Institute of Technology, BC

Project Description:



The BCIT Energy Awareness Campaign was an institute-wide initiative designed to educate and engage students, faculty, and staff in reducing energy use and greenhouse gas (GHG) emissions through behaviour change. The campaign featured events, visual displays, digital communications, and interactive quizzes focused on energy conservation, behaviour change, and fostering sustainable habits across campus to connect everyday actions with BCIT's broader energy and sustainability goals.

A key component of the campaign was the integration of Indigenous perspectives. The project team invited an Indigenous staff member to share his insights on energy conservation and environmental stewardship through an educational video. This approach deepened participants' understanding of sustainable practices and highlighted the importance of respecting the Earth in how we produce and use energy.

Key Highlights:

- **5 campuses** – activated through daily pop-up events featuring in-person engagement, physical signage, quizzes, and digital communications.
- **1,000+ students and staff** – reached through multi-channel campaign touchpoints reinforcing energy awareness messages.
- **329 pledges** – collected from students and staff committing to energy-saving habits.
- Indigenous approaches to environmental stewardship and respect for the earth were incorporated into campaign content, reinforcing values of conservation, intergenerational responsibility, and sustainable resource use.
- The campaign added a public-facing, human-centred layer to BCIT's existing investments in energy infrastructure and carbon tracking by directly engaging the campus community in day-to-day energy decisions.



BCIT is committed to continuing its efforts to raise energy-saving awareness, with the full support and active involvement of the Green Team members.





The nutrient journey:
From culinary arts to greenhouse and back

University College of the North

Project Leads: Tara Manych and Vanessa Rosenkranz
University College of the North, The Pas, MB

University College of the North

Project Description:



An novel composting program has been created at UCN The Pas campus with vermicomposting colonies established in UCN's passive solar greenhouse. As a vibrant reminder of the program's purpose, a mural artwork was created by a renowned Winnipeg artist, with contributions from local high school students. The program and artwork emphasize the stewardship role humans share for wildlife, soil, water, and air.

Project contributors have propagated the vermicompost colonies in the greenhouse and have been developing measurement protocols and additional composting methods in support of the overarching goal. Internal and external collaboration has brought this project momentum and is promising for continued growth and research.

Key Highlights:

- First-time composting program established at UCN The Pas campus.
- **75 L of organic waste** – redirected from landfill into soil production through preliminary steps.
- Activities and meetings have enabled intergenerational knowledge transfer between indigenous knowledge keepers, artists, researchers, and students in the spirit of soil and climate stewardship
- **7 stakeholder groups** – aligned through the project's activities and collaborations.
- Inspirational artwork reveal to be accompanied by words from local knowledge keepers and waste reduction researchers



Following the March 2026 Art Reveal, University College of the North will expand community engagement by integrating the composting program into campus courses and measuring its environmental impact. Further collaborations will explore scaling the model to additional northern campuses and local food networks.





Planting for the Future:
A Campus Micro-Forest Initiative

Conestoga College

Project Lead: Laura Matheson
Conestoga College, Kitchener, ON

CONESTOGA
Connect Life and Learning

Project Description:



On October 28, 2025, Conestoga College hosted a Micro-Forest planting initiative as part of its broader climate action and campus rewilding efforts. The event engaged 40 students, faculty, and staff in a three-hour hands-on planting activity, resulting in the installation of over 100 native shrubs and trees.

Beyond increasing campus biodiversity, the initiative directly contributes to greenhouse gas (GHG) reduction. Over their lifetime, the planted species are expected to sequester approximately 200–250 kg of carbon annually — roughly equivalent to the annual energy use of one household.

The planting plan was informed by Indigenous knowledge and perspectives through consultation with Indigenous community leadership. This approach ensured that species selection and ecological planning reflected local land-based knowledge systems and principles of environmental stewardship.

Key Highlights:

- A **3-hour collaborative planting event**, designed not only as an environmental intervention but also as an educational opportunity promoting ecological literacy and stewardship.
- **40 participants** from across the Conestoga community, including students, faculty, and staff, contributing to a shared hands-on climate action experience.
- **100+** native shrubs and trees planted, increasing on-campus biodiversity and supporting long-term habitat creation for local species.
- **200–250 kg of carbon** – estimated to be sequestered annually over the plants' lifetime, contributing to long-term GHG reduction goals.
- Carbon impact roughly equivalent to the annual energy use of one household, providing a relatable benchmark for understanding the project's environmental value.



Conestoga College is currently developing a comprehensive watering and long-term maintenance plan to ensure the micro-forest establishes successfully and delivers lasting ecological benefits. In the next phase, the college will explore planting a second adjacent micro-forest to expand biodiversity and double the positive environmental impact.



